Team Details

a. Team name: SAS CODIES
b. Team leader name: Saud Satopay
c. Problem Statement: Track 03 — Financial Health Score (AI/ML MSME Financial Health Card on GST · UPI · AA · EPFO)

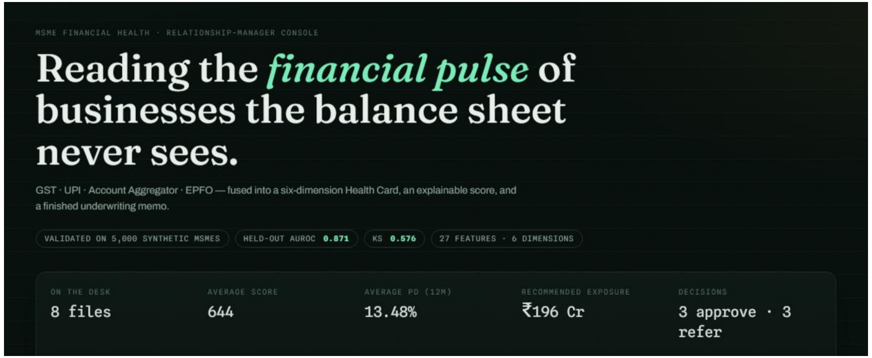
Brief about the idea

NAADI (नाड़ी — “the pulse”)

An AI-native **MSME Financial Health Card** plus **Munshi**, an agentic underwriting copilot. NAADI fuses four consented data rails — **GST, UPI, Account Aggregator, EPFO** — into a six-dimension, explainable 300–900 score, a credit decision with a limit, and a committee-ready sanction memo.

score → reasons → decision → memo → monitoring

Built for **New-to-Credit / New-to-Bank** enterprises that have no audited financials and no bureau file — the exact gap the Bank's problem statement names. Working prototype: trained model (held-out **AUROC 0.871**), live console, decision API — all public.



Opportunities

How is it different?

Six explainable dimensions + honest confidence band — not one opaque number

Monotone scorecards + TreeSHAP reason codes a credit committee can audit

Thin-file tiers widen uncertainty instead of rejecting data-poor borrowers

LLM writes prose, never numbers — deterministic engine owns every figure

How does it solve the problem?

• Scores **credit-invisible MSMEs in seconds** from consented alternate data

• Cuts application-to-decision **from days to minutes** (Munshi drafts the memo)

Rolling 12-month rescore flags stress quarters early — origination + monitoring

• Every decline ships a **what-if comeback path** (e.g. +24 pts for on-time GST ×3 mo)

USP

An underwriting brain, not a score — decision, limit, covenants, triggers included

Interrogable AI: sensitivity sliders answered by the trained model itself

Human gate built in: officer concurs or overrides on the record

ULI / OCEN / AA-native design with a working POST /score API today

List of features offered by the solution

Six-dimension Health Card — Liquidity · Stability · Growth · Repayment · Compliance · Concentration

Calibrated PD(12m) → **300–900 score** with tier-based confidence band

TreeSHAP reason codes in plain language (+ score waterfall)

What-if engine — actionable counterfactual paths for the borrower

Sensitivity lab — drag a lever, the trained model projects the score

Stress lab — adverse scenarios rescored, not guessed

Munshi memo — decision, limit, tenor, covenants, EWS triggers (Claude-drafted, engine-numbered)

Officer review gate — concur / override with recorded rationale

Portfolio radar — rolling rescore + rule-based EWS alert feed

Risk map — whole book by score × PD × exposure

⇄ **Compare** — two files on one radar with dimension deltas

Decision API — POST /score for LOS / ULI / OCEN integration

Process flow diagram or Use-case diagram

1 · CONSENT
AA consent artefact
GSTN OTP · DPDP purpose-bound
immutable audit ledger

2 · PULL RAILS
GSTR-1/3B returns
UPI txns · AA bank statements
EPFO payroll (ECR)

3 · FEATURES
27 point-in-time signals
leak-proof as-of joins
tier T1/T2/T3 coverage

4 · SIX SCORECARDS
monotone LightGBM
L·S·G·R·C·K subscores 0–100
sector-peer percentile scaled

5 · SCORE + PD
composite → isotonic PD(12m)
300–900 with ± tier band
benchmarked vs 5,000 peers

6 · EXPLAIN
TreeSHAP reason codes
what-if comeback paths
sensitivity + stress labs

7 · POLICY + LIMIT
cash-flow affordability (EMI room)
guards: L-floor · K-cap · C-gate
limit · tenor · covenants

8 · MUNSHI MEMO
Claude drafts the prose only
every figure injected by engine
evidence-linked, committee-ready

9 · OFFICER GATE
a human always decides
concur / override + rationale
recorded to audit trail

10 · MONITOR
rolling monthly rescore
EWS alerts below 580
portfolio radar feed

feedback: monthly rescore

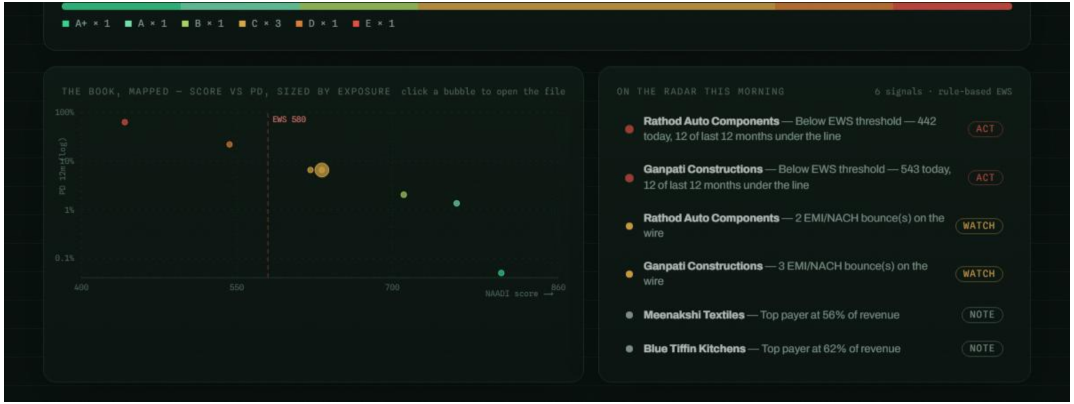
✓ **APPROVE — sanction with limit, tenor, covenants**

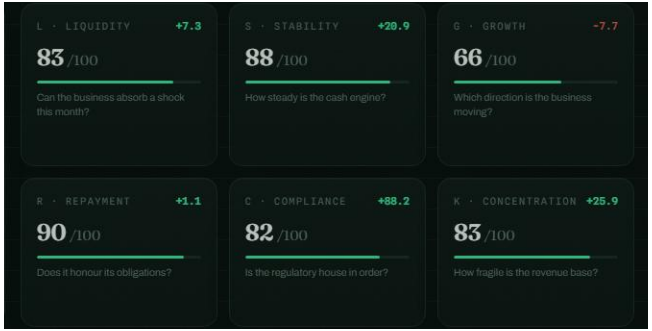
◐ **REFER — committee with mitigants**

✕ **DECLINE — with a what-if comeback plan**

Every run is consent-first and fully audited; the officer — not the model — owns the final decision.

Wireframes/Mock diagrams of the proposed solution (optional)





Live console (not a mock):

risk map of the book + EWS alert feed (left) · the six-dimension card grid (above) — all rendered from engine output.

naadi-kappa.vercel.app

Architecture diagram of the proposed solution

CONSENTED RAILS

- GSTN — GSTR-1/3B, filing discipline
- UPI — daily pulse, payer graph
- Account Aggregator — balances, bounces
- EPFO — headcount, wage trend
- Consent ledger — DPDP, immutable audit

→

THE TWO-LAYER BRAIN

- 27 point-in-time features (Polars, leak-proof)
- 6 monotone LightGBM scorecards (L·S·G·R·C·K)
- Composite → isotonic PD(12m) → 300–900 ± band
- TreeSHAP reasons · what-if · sensitivity · stress
- Peer benchmarks vs 5,000-MSME population

→

MUNSHI + SURFACES

- Deterministic policy & limit engine
- Claude-drafted memo (prose only, fact-sheet injected)
- Officer gate: concur / override, recorded
- RM console · print-ready sanction memo
- POST /score API → LOS · ULI · OCEN rails

Demo: Polars + DuckDB on laptop → Scale path: Kafka → Iceberg · Feast · MLflow · EKS · OpenTelemetry — connectors swap to IDBI sandbox with zero downstream changes.

Technologies to be used in the solution

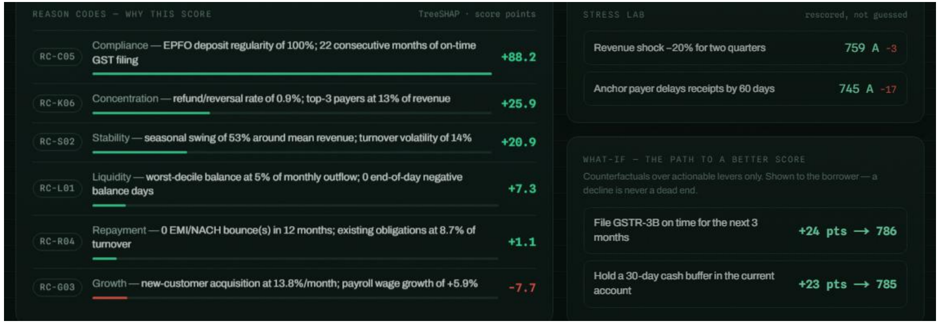
Console	Next.js 16 (App Router, Turbopack) · React 19 · Tailwind CSS v4 · Motion 12 · Recharts 3 — fully static
Engine	Python 3.13 · uv · Polars · DuckDB · scikit-learn 1.9 · FastAPI (POST /score, p95 < 300 ms)
ML	LightGBM 4.6 monotone scorecards · isotonic calibration · native TreeSHAP · counterfactual & sensitivity engines
Copilot	Claude Opus 4.8 (adaptive thinking) for memo prose · deterministic offline fallback — demo can never break
Data & consent	Synthetic latent-risk generator today → IDBI sandbox APIs (Jul 22) · AA/GSTN/EPFO connectors · DPDP-aligned consent ledger
Scale path	AWS (EKS) · Kafka → Iceberg lakehouse · Feast feature store · MLflow registry · OpenTelemetry + Langfuse

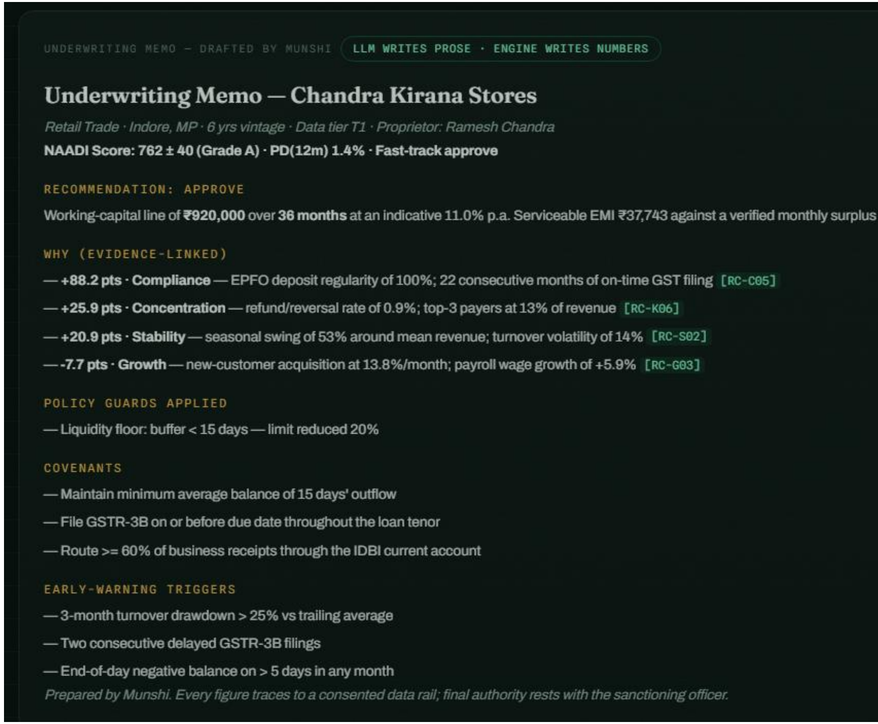
Estimated implementation cost (optional)

Concept build (done)	₹0 infrastructure — static console (Vercel), model trains on a laptop in ~40 s
Sandbox PoC (Jul 22–31)	Covered by hackathon-provided AWS credits + sandbox APIs; no bank spend
Pilot (indicative)	Serving: containerised FastAPI on 2 small nodes ≈ ₹25–40k/mo · memo LLM ≈ ₹1–3 per file (offline fallback → ₹0)
At scale (indicative)	< ₹2 marginal compute per decision at 10k decisions/day; data-rail costs per GSP/AA pulls as contracted

Cost discipline is architectural: the LLM only writes prose (small, cacheable calls); scoring is pure CPU.

Snapshots of the prototype





Score waterfall · TreeSHAP reason ledger · stress lab (left) — Munshi's evidence-linked sanction memo (right). All live at naadi-kappa.vercel.app.

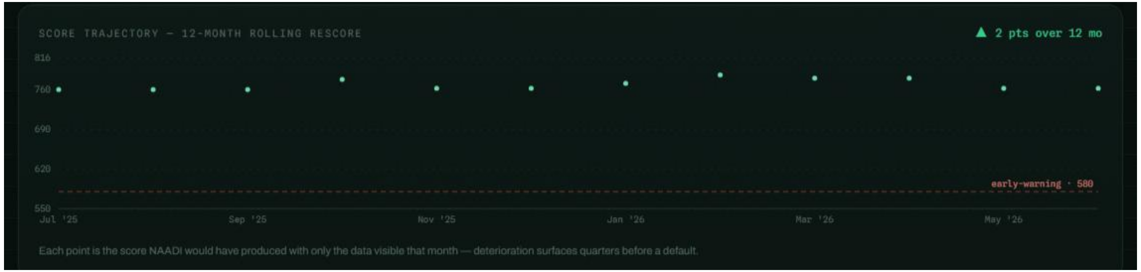
Prototype Performance report/Benchmarking

0.871
held-out AUROC — composite PD model

0.576
KS statistic (test fold)

5,000
synthetic MSME validation population

< 300 ms
p95 · POST /score decision API



Early-warning proof:

rolling rescore kept the failing supplier **below the EWS line for all 12 months** before its PD hit 63%.

Why not "90% accuracy"? Default is a rare event — accuracy is a broken yardstick. We report **AUROC / KS + calibration** and will defend that choice.

Synthetic-population validation of the architecture — not a market claim. Re-estimated on IDBI sandbox data (Jul 22–31) with a published report.

Additional Details/Future Development (if any)

Jul 22–31 · Sandbox	Retrain + isotonic recalibration on IDBI data; publish validation report (AUROC/KS/PSI, fairness slices); live-data demo
PoC · LOS integration	POST /score into loan origination; swap-set study — how many good NTC borrowers were being declined?
Rails	ULI / OCEN 4.0 embedded journeys — score at the point of commerce; AA-consent UX hardening
Borrower side	Vernacular MSME app: my Health Card, my what-if plan, my 90-day “build your NAADI” track
Monitoring	Productise the radar into a Track-04-grade early-warning system for the whole MSME book

Provide links to your:

GitHub (public)

github.com/SaudSatopay/naadi

Final product (live)

naadi-kappa.vercel.app

In-repo documentation: ARCHITECTURE.md · SCORING.md · PITCH.md · 9-page concept deck (PDF)

score → reasons → decision → memo → monitoring

NAADI — because the businesses Bharat runs on deserve to be seen.

IDBI BANK

IDBI 2026 INNOVATE

Build. Integrate. Transform.

Knowledge Partner: aws

Powered by: H2S

Technology Partner: acc applied cloud computing

Thank you!